



Pedram Nikjooy

CLOUD & DEVOPS ENGINEER

Torino, Italy · EU citizen
+39 331 272 4715
p_nikjooy@yahoo.com
linkedin.com/in/pedram-nikjooy
github.com/pedramnj
pedramnikjooy.me

TECHNICAL SKILLS

CLOUD & INFRASTRUCTURE

AWS Hetzner S3 Storage
Multi-Cloud Cloud-Native

CONTAINERS & ORCHESTRATION

Kubernetes Docker Docker Compose
k3s Helm

CI/CD, IAC & MLOPS

GitHub Actions CI/CD Pipelines IaC
MLflow MLOps Git

LANGUAGES & BACKEND

Python SQL Bash Linux
FastAPI Celery Redis

DATA & MONITORING

PostgreSQL TimescaleDB
Prometheus Grafana Alertmanager
Nginx

GENAI & ML

LLM Agents MCP Ollama / Groq
Gradient Boosting NumPy / Pandas

EDUCATION

M.Sc. Computer Engineering
Cloud Computing

Politecnico di Torino

2022 – Jul 2026 · Torino, Italy

Thesis: *AutoSage* — LLM-driven predictive autoscaler for Kubernetes.

B.Sc. Software Engineering
IAU-TNB

2014 – 2018 · Tehran, Iran

CERTIFICATIONS

Meta Front-End Developer
Meta · 2024

Technical Support Fundamentals
Google · 2024

LANGUAGES

Italian B1
English C2
Persian Native

PROFESSIONAL SUMMARY

M.Sc. Computer Engineering (Cloud Computing) graduate from Politecnico di Torino (July 2026), focused on cloud-native infrastructure, Kubernetes, and applied AI. Built **AutoSage** for the master's thesis — an LLM-driven predictive autoscaler that cut cost per SLA-met request by 28–47% versus native Kubernetes autoscaling. Founder and technical lead of **STRYX**, an AI structural-monitoring startup incubated at Politecnico's I3P, and builder of production-grade ML platforms end to end. Hands-on with Docker, Kubernetes, CI/CD, MLOps, and GenAI agents, with practical Linux and cloud-infrastructure experience. Eager to help design and deliver multi-cloud enterprise architectures accelerated by GenAI, Infrastructure as Code, and advanced MLOps pipelines in fast-paced consulting and energy-sector contexts.

MASTER'S THESIS

AutoSage — LLM-Driven Predictive Autoscaler for Kubernetes 2025 – 2026
Politecnico di Torino · M.Sc. Thesis

- ▶ Built a production-style Kubernetes autoscaler that combines **LLM reasoning** with TOPSIS multi-criteria decision analysis to choose between horizontal (HPA) and vertical (VPA) scaling on live clusters.
- ▶ Engineered a fault-tolerant **LLM cascade** (local Qwen via Ollama → Groq cloud fallback → deterministic rules) and an **MCP-based service stack** (FastAPI server, Flask-SocketIO UI, k3s integration).
- ▶ Benchmarked against native HPA, VPA, and an RL baseline across **240+ trials** — cut **cost per SLA-met request by 28–47%** vs. HPA while respecting scaling best practices by workload type.
- ▶ Stack: *Python, Kubernetes / k3s, Ollama, Groq, MCP, FastAPI, Flask-SocketIO, NumPy / Pandas.*

PROJECTS & VENTURES

STRYX — Founder & Technical Lead 2025 – Present
I3P Incubator · Politecnico di Torino

- ▶ Founded and lead the technical build of **STRYX**, an AI platform that turns instrumented buildings into autonomous structural inspectors — classifying each as **Safe / Restricted / Unsafe** within minutes of a seismic event.
- ▶ Architected and operate the full **cloud-native stack**: containerised FastAPI microservices, PostgreSQL / TimescaleDB, Celery / Redis pipelines, and S3-compatible storage, orchestrated with Docker and automated CI/CD.
- ▶ Built a production **ML pipeline** (gradient-boosted classification with conformal-prediction uncertainty and anomaly detection) reaching **0.97 macro-F1**.
- ▶ Advancing through the **I3P incubation programme** toward pilot deployment and seed funding.

PeDax — ML Crypto-Trading Platform

2026 – Present

Independent Project

- ▶ Designed and built **end to end** a production-hardened, multi-symbol trading platform: real-time market-data ingest, walk-forward ML with calibrated probabilities, a per-asset risk engine, and paper / live execution with shadow validation.
- ▶ Built a **Next.js** operator dashboard with live SSE streams and offline queue; deployed on **Hetzner** with Docker Compose, **GitHub Actions CI/CD**, and Prometheus / Grafana / Alertmanager monitoring.
- ▶ Ran pre-registered experiments through a **CV** → **shadow** → **promotion** harness and engineered risk guardrails (exposure caps, regime gating, idempotent OMS) for a realized-PnL evaluation track.
- ▶ *Stack: Python / FastAPI, Next.js 15, PostgreSQL, Redis, Docker Compose, GitHub Actions, MLflow.*

PROFESSIONAL EXPERIENCE

Technical Support Engineer – Hosting (L1 → L2)

Oct 2019 – Dec 2021

IRANHOST · Tehran

- ▶ Administered and monitored Linux web / hosting servers, ensuring **high availability** through proactive maintenance, backups, and resource optimisation.
- ▶ Configured DNS, email, and web services and resolved complex server-side and performance issues across multi-tenant environments.
- ▶ Implemented **security protocols, patching, and audits** to harden infrastructure and reduce vulnerabilities.